

Harshavardhan Hajeri

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Education

Birla Institute of Technology and Science, Pilani

Oct 2021 – June 2026

M.Sc. (Hons.) Physics | B.E. Mechanical Engineering (Dual Degree)

Experience

Research Staff | Indian Institute of Technology, Madras

June 2026 – Present

- Building a cascaded bow-tie SHG→OPO setup under Prof. Anil Prabhakar to generate and characterize squeezed vacuum; locking the bow-tie SHG cavity (PPLN, 1064 → 532 nm) that pumps the OPO via **Pound-Drever-Hall (PDH)** feedback through a Red Pitaya lock-in/PID controller.

Thesis Student | Indian Institute of Technology, Madras

June 2025 – June 2026

- Completed two M.Sc. theses under Prof. Anil Prabhakar (co-supervised by Prof. Sanjib Dey, BITS Goa); built a **Spatial Photonic Ising Machine** on a phase-only SLM with Metropolis simulated annealing and Gaussian beam compensation, and developed **holographic modal decomposition** of a bow-tie SHG cavity, raising average modal self-overlap from 0.86 to 0.96 (fundamental mode: 0.996).
- Aligned and **PDH-locked** a triangular mode-cleaner cavity at 1064 nm (99% mode matching), building the locking chain – a Red Pitaya lock-in/PID controller driving a custom inverting HV amplifier into the cavity PZT.

Research Intern | CeNSE, Indian Institute of Science, Bengaluru

June – August 2024

- Designed a cryostat dipstick in Fusion 360 for low-temperature transport measurements and built a PyQt5/QCoDeS GUI for automated I-V characterization under Dr. Dhavala Suri.

Quantum Computing Lead | Google Developer Student Club, BITS Goa

Aug 2023 – May 2025

- Led workshops and study groups reaching 100+ students; organized IBM-sponsored Qiskit Fall Fest 2023 and received the 2024 Qiskit Fall Fest Mentor Badge.

Projects

Discrete-Time Quantum Walk | *Qiskit*

- Implemented coined and split-step DTQW circuits on Qiskit for state preparation and RL-based optimization; demonstrated advantage over classical random walk baselines on the N-Armed Bandit problem.

Entanglement in Cluster States | *Strawberry Fields*

under Dr. Sanjib Dey, BITS Goa

- Simulated 2D cluster states and computed entanglement measures on photon-subtracted squeezed vacuum states in continuous-variable quantum optics.

MEMS Microphone Simulation | *COMSOL Multiphysics*

under Dr. D. G. Patil, BITS Goa

- Performed structural-acoustic coupled simulation of a capacitive MEMS microphone; proposed design modifications that improved simulated sensitivity.

Skills

Programming & Frameworks: Python, Qiskit, QuTiP, PennyLane, Strawberry Fields, JavaScript (React), L^AT_EX

Tools & Platforms: Git/GitHub, MATLAB, LabVIEW, Simulink, COMSOL, FINESSE 3, Red Pitaya, IBM Quantum, Fusion 360, Jupyter Notebook

Teaching Experience

Teaching Assistant | *Quantum Information & Computation, BITS-F386*

Jan – May 2024

- Under Dr. Radhika Vathsan - designed and graded assignments for 50 students; created Qiskit-based course content and Jupyter Notebook tutorials.

Mentor | *Quantum Computing with Qiskit (Quark STP) | Theory of Relativity (CTE)*

2022 – 2023

- Mentored 150+ students across two programmes; developed Manim-based materials covering Grover's Algorithm and the foundations of special relativity.

Conference Presentations

G. Patil*, **H. Hajeri***, S. P. Amrithraj, A. Prabhakar. *Modal Decomposition of Cavity SHG Fields using Spatial Light Modulator*. EOP 2025.

H. Hajeri*, N. Vinod P.M., G. Patil, S. P. Amrithraj, A. Prabhakar. *Spatial Photonic Ising Machine using Spatial Light Modulators*. EOP 2025.

Awards & Extra-Curriculars

Seed Fund Grant INR 5,00,000 (PIEDS) | Prof. Suresh Ramaswamy Award INR 40,000 | IBM Qiskit Fall Fest Mentor Badge, 2024

Steering Member, QIndia | Core Member, Nirmaan NGO, BITS Goa | Core Member, Mime Club BITS Goa